Diameter-7 cells from a spread and its whiskers: the freed sets and the bounds for every q

Degree–diameter lab, 29 October 2026 (job thm-conics)

Abstract. The jobs "conics" and "conics-2" raised the lab's values at the diameter-7 cells $(14, 7)$ to $(20, 7)$ by freeing, on the friendly transitions of the typed hexagon layouts, two sets: the points of the Hermitian spread S of $H(q)$, and the lines meeting a spread line v together with an exact whisker set and the q^2 spread lines that carry no whisker. Here these steps are turned into statements for every prime power q. (1) For any generalized hexagon of order (q, q), any spread S, any $v \in S$ and any whisker set W, the lines meeting v, the whiskers and the spread lines missed by the whiskers form a dual segment set; for the exact whisker set of `POLISHED.md` (every odd q and every even $q \geq 4$) it has exactly $q^4 + q^2 + q$ lines and is maximal, while the spread points form a segment set of $(q+1)(q^3+1)$ points. (2) For $2k+1$ levels, any regular tournament and any typing, some level has at most $\lfloor k/2 \rfloor$ friendly outgoing transitions and some level at most $\lfloor k/2 \rfloor$ friendly incoming ones; this bounds the hub counts, with the exact second-order term, and the alternating typing attains the bound for the total number of hubs for every k. (3) With the $q^2 + 1$ points of an elliptic quadric section on the non-friendly transitions, this gives, for every $k \geq 2$ and every prime power $q \geq q_k$,

$$n_b(q + 1 + k, 7) \;\geq\; 2(2k + 1)(q + 1)(q^4 + q^2 + 1) + \begin{cases} k\,(q^4 + 2q^2 + q + 1) & k \text{ even}, \\ \frac{k-1}{2}\,(2q^4 + q^3 + q^2 + 2q + 1) + (k + 1)(q^2 + 1) & k \text{ odd}, \end{cases}$$

with $q_2 = 16$, $q_3 = q_4 = 23$, $q_5 = 29$, $q_k \leq 300k$ in general. The hub graphs are explicit (sums of blown-up hexagons); smaller q are checked by computer where possible. The formulas reproduce the vertex counts of all fifteen graphs built in the two jobs. At $(19, 7)$ the formula equals the built value 11,316,940 and every ingredient is proved. At $(18, 7)$ the theorem with an explicit hub graph (two copies of the incidence graph of $H(7)$ joined by perfect matchings) gives 7,318,644, above the value 7,309,092 of branch `claude/conics-2-diameter-7-ic0oat` (not of master). Every ingredient is explicit, so no computer search is needed for this bound; the graph was also built and checked from every hub (§5.1).

Labels. Every result is marked **Theorem** (proved here or in the cited note), **Computer-checked** (a program checked it; the program is in the repository) or **Heuristic**, and **[new]** or **[restatement]**. "New" means new in the lab's notes; the literature check of `D7_HEXAGON.md` §11 applies, and we did not search the literature again. **[standard]** marks a known result, with its source given.

Sources. `research/synthetic/CONICS.md` §3.4 (branch `claude/conics-generalized-polygons-6gjqxi`); `research/synthetic/conics2/README.md` (branch `claude/conics-2-diameter-7-ic0oat`, state of 11:44 UTC); `research/theorems/D7_HEXAGON.md` (branch `claude/d7-hexagon-theorems-4w1prt`), whose Definitions 2.1, 3.1, Theorems A, B, C, 3.4 and Proposition 5.2 are used throughout; `research/theorems/POLISHED.md` §6 (branch `claude/thm-polish-2oddll`); `research/synthetic/SYNTHETIC.md` (branch `claude/synthetic-incidence-geometry-s4m08f`), whose conventions we follow. Programs of this note: `research/synthetic/d7thm/`; squad round `research/squads/d7thm-r1/`.

0. Setting

Hexagons. A generalized hexagon Γ of order (q, q) has $N = N(q) = (q + 1)(q^4 + q^2 + 1)$ points and as many lines; $q + 1$ points on a line and $q + 1$ lines on a point. Distances are taken in the incidence graph I (diameter 6, girth 12); elements at distance 6 are *opposite*. For an element x, $|\Gamma_i(x)| = (q + 1)q^{i-1}$ for $1 \leq i \leq 5$. A *spread* is a set of $q^3 + 1$ pairwise opposite lines; every line not in a spread S meets exactly one line of S, in one point

(D7_HEXAGON.md, Lemma 1.2(c)). The example throughout is the split Cayley hexagon $H(q)$ in $Q(6,q)$ and its Hermitian spread: the lines of $H(q)$ inside a hyperplane Π with $\Pi \cap Q(6,q)$ elliptic (every q; [Th80]).

Segment sets (D7_HEXAGON.md, Definition 1.3). A point set F is a *segment set* if for every $p \in F$ the points of $F \setminus \{p\}$ collinear with p lie on one line through p. A line set G is a *dual segment set* if for every $\ell \in G$ the lines of $G \setminus \{\ell\}$ that meet ℓ pass through one point of ℓ. Both kinds have at most $(q+1)(q^3+1)$ elements (D7_HEXAGON.md, Proposition 1.4).

Layouts. $L = 2k+1$ *levels*, each a copy of I of *type* H (colour $\mathcal{A}$ = points, $\mathcal{B}$ = lines) or D (colour $\mathcal{A}$ = lines, $\mathcal{B}$ = points). The levels are the vertices of a *regular tournament* T: every two levels are joined by exactly one *transition* $j \to j'$, and every level has k outgoing and k incoming transitions. Regularity is also what makes the degree $q+1+k$: a level with $k+1$ outgoing transitions would give its $\mathcal{A}$-elements degree $q+k+2$, so regular tournaments are the class of Theorem 2.1. Here k is the number of transitions out of a level, which D7_HEXAGON.md calls r; there k is the step of a transition (j,k). A transition is a bijection from the $\mathcal{A}$-elements of level j to the $\mathcal{B}$-elements of level j'; it is *friendly* if the types differ, and then it is the identity. D7_HEXAGON.md uses the rotational tournament $j \to j+i$ ($1 \le i \le k$, indices mod L); Theorem A and Theorem 3.4 there hold verbatim for every regular tournament, since their proofs only use that every pair of levels is the pair of one transition. The *freed set* R_t of a transition is the set of $\mathcal{A}$-elements whose matching edge is deleted. Level j then has $\alpha_j = \sum_{t \text{ out of } j} |R_t|$ free $\mathcal{A}$-slots and $\beta_j = \sum_{t \text{ into } j} |R_t|$ free $\mathcal{B}$-slots. $K_{\mathcal{B}} \le \min_j \alpha_j$ hubs of colour $\mathcal{B}$ each take one free $\mathcal{A}$-slot in every level, $K_{\mathcal{A}} \le \min_j \beta_j$ hubs of colour $\mathcal{A}$ one free $\mathcal{B}$-slot in every level, and a bipartite *hub graph* Y joins the two kinds of hubs.

What repairs a transition. Theorem B of D7_HEXAGON.md: a friendly transition is repaired if its freed set is a segment set (source of type H) or a dual segment set (source of type D). Theorem C there, with Corollary 5.3(a): a non-friendly transition $H \to H$ with matching σ is repaired by a set R of m pairwise opposite points when σ maps R onto m pairwise opposite lines and the points collinear with x onto the lines concurrent with $\sigma(x)$ for every $x \in R$; such a σ exists for every partial ovoid R and every partial spread of the same size. A $D \to D$ transition uses σ^{-1} and frees $\sigma(R)$.

Hub graphs. $\mathrm{HG}(K_1, K_2; \delta)$ denotes a bipartite graph with colour classes of sizes K_1 and K_2, maximum degree at most δ, in which any two vertices *of the same colour* are at distance at most 6.

Theorem 0.1 (D7_HEXAGON.md, Theorem 3.4) [restatement]. If every transition is repaired, $K_{\mathcal{B}} \le \min \alpha_j$, $K_{\mathcal{A}} \le \min \beta_j$ and the hub graph is an $\mathrm{HG}(K_{\mathcal{B}}, K_{\mathcal{A}}; q-k)$, then the graph has maximum degree $q+1+k$ and diameter at most 7, and

$$n_b(q+1+k, 7) \;\ge\; 2(2k+1)N(q) + K_{\mathcal{A}} + K_{\mathcal{B}}.$$

The value $2(2k+1)N(q)$ alone (no freed sets, no hubs) is the value of the 2013 table at the cells $(7,7)$, $(9,7)$–$(16,7)$ [MS13] for the pairs $(q,k) = (5,1), (7,1), (8,1), (9,1), (9,2), (11,1), (11,2), (11,3), (13,2)$ (cells $\Delta = q+1+k = 7, 9, 10, \ldots, 16$ in this order), and the lab's beyond-table value at $(17,7)$–$(20,7)$.

What uses coordinates. Sections 2 and 4 are about tournaments and bipartite graphs only. In §1, Theorem 1.4, Proposition 1.5 and Lemma 1.6 hold in every generalized hexagon of order (q,q) with a spread; only Theorem 1.7 (the whisker set of POLISHED.md meets each spread line in 0 or $q+1$ points, so exactly q^2 spread lines are missed) and the existence statements (Hermitian spread, exact whisker set, elliptic quadric section) use the coordinates of $H(q)$. Coordinates appear only in proofs.

Notation for sizes. $P(q) = (q+1)(q^3+1) = q^4+q^3+q+1$, $\Lambda(q) = q^4+q^2+q$, and m the size of the non-friendly sets. Note $P - \Lambda = q^3 - q^2 + 1$.

1. The freed sets

1.1 The configuration

Definition 1.1. Let S be a spread of Γ, $\mathcal{P}_S$ the set of points on its lines, and $v \in S$. $\mathcal{L}_v$ is the set of the $q(q+1)$ lines other than v that meet v. A point is *free* if it is at distance 5 from v and not in $\mathcal{P}_S$. A *whisker set* is a set W of lines, each opposite v and not in S, such that no free point lies on two lines of W; it is *exact* if every free

point lies on exactly one line of W. Finally

$$S_0(W) = \{n \in S \setminus \{v\} : \ n \text{ meets no line of } W\}, \qquad R_L(W) = \mathcal{L}_v \cup W \cup S_0(W).$$

(This is the configuration of Theorem D, `POLISHED.md` §6.2 and `SYNTHETIC.md` §3.2, written without coordinates.)

Lemma 1.2 [restatement, with the counts, parts (c)–(e) from `POLISHED.md` §6.2; Theorem]. (a) The points at distance at most 3 from v are the $q+1$ points of v and the $q^2(q+1)$ points at distance 3, each of which lies on exactly one line of $\mathcal{L}_v$. (b) The points of $\mathcal{P}_S$ not on v are at distance 5 from v. (c) There are $q^3(q^2-1)$ free points. (d) A line opposite v and not in S has exactly one point in $\mathcal{P}_S$ and q free points. (e) A whisker set has at most $q^2(q^2-1)$ lines, with equality if and only if it is exact.

Proof. (a) A point at distance 3 from v lies on a line meeting v; two such lines through it would close a circuit of length 4 or 6. Counting: $|\Gamma_3(v)| = (q+1)q^2$. (b) A point x on a spread line $n \neq v$ has $d(x,v)$ odd, at most 5 and at least $d(n,v)-1 = 5$. (c) $|\Gamma_5(v)| = (q+1)q^4$, of which $q^3(q+1)$ lie in $\mathcal{P}_S$ by (b). (d) Every point of a line opposite v is at distance 5 from v; the line meets exactly one line of S, in one point. (e) By (d) and (c), counting incidences between free points and W. □

1.2 Spread points and whiskers

Proposition 1.3 (`D7_HEXAGON.md`, Proposition 1.4) [restatement; Theorem]. $\mathcal{P}_S$ is a segment set of $P(q) = (q+1)(q^3+1)$ points, and no segment set is larger; equality holds only for the point sets of spreads.

Theorem 1.4 [new in this form; Theorem]. Let S be a spread, $v \in S$ and W any whisker set.

(a) $R_L(W) = \mathcal{L}_v \cup W \cup S_0(W)$ is a dual segment set, and two of its lines that meet, meet in a point of $\mathcal{P}_S$.

(b) $|R_L(W)| = q(q+1) + |W| + |S_0(W)|$.

(c) If W is exact and every point of $\mathcal{P}_S$ not on v lies on a line of W or of $S_0(W)$, then every point of Γ lies on a line of $R_L(W)$, and no line can be added to $R_L(W)$: it is a maximal dual segment set.

(d) A line $n \in S \setminus \{v\}$ all of whose whiskers pass through one point of n can be added to $R_L(W)$ as well.

Proof. (a) A line of $\mathcal{L}_v$ has one point on $v \subseteq \mathcal{P}_S$ and its other q points at distance 3 from v, so not in $\mathcal{P}_S$ (Lemma 1.2(b)). A whisker has one point in $\mathcal{P}_S$ and q free points (Lemma 1.2(d)). So every line of $\mathcal{L}_v \cup W$ has exactly one point in $\mathcal{P}_S$. Its other points are either at distance 3 from v, and then on no other line of $\mathcal{L}_v$ (Lemma 1.2(a)) and on no whisker (whose points are at distance 5), or free, and then on no line of $\mathcal{L}_v$ and on at most one whisker. So two lines of $\mathcal{L}_v \cup W$ never meet outside $\mathcal{P}_S$. A line $n \in S_0(W)$ meets no line of $\mathcal{L}_v$ (whose points are at distance at most 3 from v, while those of n are at distance 5), no whisker (definition) and no other spread line. So the lines of $S_0(W)$ are isolated, and for every other line ℓ of $R_L(W)$ the lines of $R_L(W)$ meeting ℓ all pass through the unique point of ℓ in $\mathcal{P}_S$. (b) The three sets are disjoint: whiskers are opposite v, lines of $\mathcal{L}_v$ are not, and neither kind is in S. (c) Points of v and points at distance 3 from v lie on lines of $\mathcal{L}_v$, free points on whiskers (W exact), and the points of $\mathcal{P}_S$ off v by hypothesis. A line $\ell \notin R_L(W)$ then meets a line of $R_L(W)$ at each of its $q+1$ points, and these lines are different from each other (two points of ℓ on one line $\neq \ell$ is impossible), so $R_L(W) \cup \{\ell\}$ is not a dual segment set. (d) Such an n meets lines of $R_L(W)$ only in that point x (it meets no line of $\mathcal{L}_v$ and no other spread line), and for a whisker through x the new line meets it at its point in $\mathcal{P}_S$, where all its other meetings already are. Two such spread lines do not meet. □

For W exact and $S_0(W)$ left out this is `CONICS.md`, Lemma 3.1; for the whisker set of `POLISHED.md` and its q^2 empty spread lines it is Proposition 1(c),(d) of `conics2/README.md`. The proof above uses only incidences.

Proposition 1.5 [new; Theorem] (*where the room is*). Let G be a dual segment set in which any two lines that meet do so in a point of $\mathcal{P}_S$ (as in Theorem 1.4(a)). Then $|G \setminus S| \leq q^4 + q$, with equality if and only if the lines of $G \setminus S$ cover every point outside $\mathcal{P}_S$ exactly once. Hence $|G| \leq q^4 + q + |G \cap S|$. For an exact whisker set, $\mathcal{L}_v \cup W$ has exactly $q^4 + q$ lines.

Proof. A line not in S meets exactly one spread line, in one point, so it has exactly one point in $\mathcal{P}_S$ (a second point would lie on a second spread line). Its other q points lie outside $\mathcal{P}_S$, and by hypothesis these q-sets are disjoint for different lines of $G \setminus S$. There are $N - P(q) = q^5 + q^2 = q(q^4+q)$ points outside $\mathcal{P}_S$. Finally $q(q+1) + q^2(q^2-1) = q^4 + q$. □

So within this class $R_L(W)$ can only grow by containing more spread lines, at most q^3-q^2+1 more; the general bound $(q+1)(q^3+1) = q^4+q+(q^3+1)$ is the case where all spread lines are in G, which forces an ovoid.

Lemma 1.6 [new; Theorem]. Let W be exact. Every whisker meets exactly one line of $S\ \{v\}$, and a line $n \in S\ \{v\}$ meets at most q^2-1 whiskers. So at least q^2 spread lines other than v meet a whisker and $|S_0(W)| \le q^3-q^2$. If every line of $S\ \ \{v\}$ meets either no whisker or $q+1$ whiskers, then $|S_0(W)| = q^2$.

Proof. A whisker meets one line of S (§0), not v, since it is opposite v. At a point x of n the $q+1$ lines are n, the line through x on the geodesic to v (at distance 4 from v, so not a whisker) and $q-1$ others. Hence n meets at most $(q+1)(q-1)$ whiskers, and $q^2(q^2-1)/(q^2-1) = q^2$. In the last case the lines meeting whiskers number $q^2(q^2-1)/(q+1) = q^2(q-1)$, and $q^3-q^2(q-1) = q^2$. □

1.3 The exact whisker set of `POLISHED.md`

In $H(q)$ with the coordinates of `POLISHED.md` §6.1 (q odd) and §6.5 (q even): Π the elliptic hyperplane $X_0+cX_4 = 0$ (q odd, $-c$ a non-square) or $X_3 = X_0+\delta X_4$ (q even, $\mathrm{Tr}\,\delta = 1$), S its Hermitian spread, $v = \langle e_1, e_6\rangle$, the hyperplanes $\Pi_z \supset v$ ($z = (\alpha,\beta) \in \mathbb{F}^2$) and their spreads S_z, the function ν on the points of Π off v, the zero-free cubic G and $F(z) = G(z)/N(z)$ with N the anisotropic norm form; $W = \{\ell \in S_z\ \ \{v\} : z \ne 0,\ \nu(\ell \cap \Pi) = F(z)\}$ is exact for every odd q (Theorem 6.5) and every even $q \ge 4$ (Theorem 6.14).

Theorem 1.7 (`conics2/README.md`, Proposition 1(a),(b)) [restatement with proof; Theorem]. Let q be odd, or even with $q \ge 4$, and W the exact whisker set above. Then every point X of $\mathcal{P}_S$ not on v lies on exactly one whisker if $\nu(X) \ne 0$ and on none if $\nu(X) = 0$; ν is constant on every line of $S\ \ \{v\}$. Hence every line of $S\ \ \{v\}$ meets either $q+1$ whiskers, one at each of its points, or none, and $S_0(W) = \{n \in S\ \ \{v\} : \nu(n) = 0\}$ has exactly q^2 lines.

Proof. Whiskers through X are whiskers whose point in Π is X (their other points are off Π). On Π the equation of Π_z reduces to $\alpha X_2+\beta X_5 = 0$ (both parities), and $(X_2, X_5) \ne (0,0)$ because X is not on v (`POLISHED.md`, Lemma 6.1(d) / 6.10). So the $z \ne 0$ with $X \in \Pi_z$ form the punctured line $\{tz_0 : t \ne 0\}$, $z_0 = (X_5, -X_2)$. For each of them exactly one line of S_z passes through X (Lemma 6.1(b); for even q read Lemma 6.10 for 6.1 throughout); it is not v, it is not in S (6.1(e)), so it meets Π only in X (6.1(c)), and different z give different lines (6.1(e)). It is a whisker if and only if $\nu(X) = F(tz_0)$. As G is a cubic form and N a quadratic form, $F(tz_0) = tF(z_0)$, and $F(z_0) \ne 0$ since G has no zero. So $t \mapsto F(tz_0)$ is a bijection of $\mathbb{F}^\times$: one whisker through X if $\nu(X) \ne 0$, none if $\nu(X) = 0$. ν is constant on spread lines by `POLISHED.md`, Lemma 6.6(d) / 6.12(d). The count is Lemma 1.6. □

Corollary 1.8 [restatement; Theorem]. For every prime power q with q odd or $q \ge 4$, in $H(q)$: $R_H = \mathcal{P}_S$ is a segment set with $P(q) = q^4+q^3+q+1$ points, and $R_L = \mathcal{L}_v \cup W \cup S_0(W)$ is a maximal dual segment set with $\Lambda(q) = q^4+q^2+q$ lines (Theorem 1.4 with Theorem 1.7). In the notation of Proposition 1.5, $\mathcal{L}_v \cup W$ is extremal and $|R_L \cap S| = q^2$.

Computer-checked. `conics2/typed_build.py` asserts every statement of Theorem 1.7 and Corollary 1.8 for $q = 11, 13, 16$ inside the builds (the make logs on that branch; `conics2/README.md` also reports runs at $q = 4, 5, 7$, which are not claimed here, as no log is kept). Independently, squad round `d7thm-r1` (own code, not the builder): card c1 for $q = 3, 5, 7$ (hexagon built from the quadric and Tits' equations), card c2 for $q = 9, 11$, card c3 for $q = 2, 4, 8$: exactness of W, ν constant on spread lines, 0 or $q+1$ whiskers per spread line, $|S_0| = q^2$ equal to the lines with $\nu = 0$, the dual segment property, $|R_L| = q^4+q^2+q$ (6,651 at $q = 9$, 14,773 at $q = 11$, 276 at $q = 4$, 4,168 at $q = 8$), maximality (every outside line tested), and the segment property of $\mathcal{P}_S$; card c2 also for every admissible c at $q = 9, 11$, card c3 for every admissible (c_0, c_1) at $q = 4, 8$; negative controls fail as they should (`research/squads/d7thm-r1/scoreboard.jsonl`).

Remarks. (i) $q = 2$: Lemma 6.11 of `POLISHED.md` gives no zero-free G (card c3 re-checked all four pairs), but an exact whisker set of 12 lines exists (`POLISHED.md`, Corollary 6.15), so Theorem 1.4 gives $|R_L| \ge 18$; the largest dual segment set of $H(2)$ has 25 lines (**Computer-checked**, `conics2/README.md` §3: SAT and MaxSAT, squad round `conics2-r1` cards c6, c7). (ii) R_L is not maximum in general: $H(4)$ has a dual segment set of 288 $>$ 276 lines (**Computer-checked**, same place: card c8, re-checked by `research/squads/conics2-r1/tools/dualseg_check.py`; it is not of the kind in Proposition 1.5). Whether q^4+q^2+q can be beaten for $q \ge 5$ is open; by Proposition 1.5 a set of this kind would need more than q^2 spread lines. (iii) For $q = 3^h$ the pencils of an ovoid give a dual segment set of $P(q)$ lines, and the duality layout of `D7_HEXAGON.md`, Corollary 8.2 (every freed set of size $P(q)$) is at least as good as any typed layout: every

freed set allowed by Theorems B and C has at most $P(q)$ elements (Proposition 1.4 for segment sets; a set of pairwise opposite points, or of pairwise opposite lines, is non-collinear, resp. non-concurrent, and has at most $q^4+q^2+1 < P(q)$ elements by Hoffman's bound for the collinearity graph, whose least eigenvalue is $-(q+1)$), so $\alpha_j, \beta_j \le kP(q)$ for every typed layout, and the duality layout attains this. (iv) We know no coordinate-free description of the q^2 lines with $\nu = 0$.

1.4 The non-friendly sets

Proposition 1.9 (`D7_HEXAGON.md`, Lemma 1.2(a), Corollaries 5.3(a) and 9.1) [restatement; Theorem]. For every prime power q, the q^2+1 points of an elliptic quadric section $Q^-(3,q)$ of $Q(6,q)$ are pairwise opposite in $H(q)$, and with any q^2+1 lines of the Hermitian spread they give a local duality σ as in Theorem C. So $m = q^2+1$ is available on every non-friendly transition.

The lab's builds use larger non-friendly sets found by computer at $q = 11$ ($m = 768$, mirrored set) and $q = 13$ ($m = 792$), and $m = q^2+1 = 257$ at $q = 16$ (**Computer-checked**: the builder re-checks non-collinearity, non-concurrency and the local duality at every point).

2. Slot counts

Let T be a regular tournament on $2k+1$ levels with a typing. For a level j let f_j be the number of friendly transitions out of j and g_j the number into j. With friendly sets of sizes P (points, source H) and Λ (lines, source D) and non-friendly sets of size m,

$$\alpha_j = f_j X_j + (k-f_j)m, \quad X_j = \begin{cases} P & \tau_j = H \\ \Lambda & \tau_j = D \end{cases}, \qquad \beta_j = g_j Y_j + (k-g_j)m, \quad Y_j = \begin{cases} \Lambda & \tau_j = H \\ P & \tau_j = D \end{cases}$$

(a friendly transition into an H level comes from a D level, which frees lines). Put $a = \lfloor k/2 \rfloor$ and assume

$$P \;\ge\; \Lambda \;\ge\; m, \qquad \Lambda - m \;\ge\; a\,(P-\Lambda). \tag{$*$}$$

For the sets of §1, $(*)$ holds whenever $k \le 2q+1$ and $m \le q^2+1$, since then $a(q^3-q^2+1) \le q^4-q^3+q \le q^4+q-1$; at $q = 11$, $m = 768$ it holds for $k \le 23$, at $q = 13$, $m = 792$ for $k \le 27$. The bound needs $k \le q-1$ anyway (hub degree $q-k \ge 1$).

Theorem 2.1 [new; Theorem]. For every regular tournament on $2k+1$ levels and every typing, with $a = \lfloor k/2 \rfloor$:

(a) $\min_j f_j \le a$ and $\min_j g_j \le a$;

(b) if k is even, it is impossible that $f_j \ge a$ at every level and $f_j \ge a+1$ at every D level; likewise with g and H levels;

(c) if k is odd, it is impossible that $f_j \ge a+1$ at every D level and $g_j \ge a+1$ at every H level.

Consequently, if $P \ge \Lambda \ge m$: for k odd, $\min\alpha, \min\beta \le aP+(a+1)m$ and $\min\alpha + \min\beta \le a(P+\Lambda)+2(a+1)m$; for k even and under $(*)$, $\min\alpha \le a(\Lambda+m)$ and $\min\beta \le a(\Lambda+m)$.

Proof. Let h and d be the numbers of H and D levels. The $k-f_j$ non-friendly transitions out of an H level go to H levels, so $\sum_{\tau_j=H}(k-f_j) = \binom{h}{2}$, and likewise $\sum_{\tau_j=D}(k-f_j) = \binom{d}{2}$ and the same with g.

(a) If $f_j \ge a+1$ everywhere, then $\binom{h}{2} \le h(k-a-1)$ and $\binom{d}{2} \le d(k-a-1)$, so $h+d \le 4k-4a-2 \le 2k$, less than $2k+1$. For g the same.

(b) $k = 2a$. From $f_j \ge a$ on H levels, $\binom{h}{2} \le ha$, so $h \le 2a+1$; from $f_j \ge a+1$ on D levels, $\binom{d}{2} \le d(a-1)$, so $d \le 2a-1$. Then $h+d \le 4a < 4a+1$. For g exchange the roles of H and D.

(c) $k = 2a+1$. On H levels $g_j \ge a+1$ gives $\binom{h}{2} \le ha$ (count non-friendly incoming transitions), so $h \le 2a+1$; on D levels $f_j \ge a+1$ gives $\binom{d}{2} \le da$, so $d \le 2a+1$. Then $h+d \le 4a+2 < 4a+3$.

Consequences. A level with $f_j \le a-1$ has $\alpha_j \le (a-1)P+(a+1)m \le a\Lambda+am$, by $(*)$ (only $\Lambda-m \ge (a-1)(P-\Lambda)$ is used, which always holds for $k = 2$). If k is even and $\min\alpha > a(\Lambda+m)$, every level has $f_j \ge a$, and every D

level $f_j \geq a+1$ (a D level with $f_j = a$ has $\alpha_j = a\Lambda + am$), contradicting (b). If k is odd, a level with $f_j \leq a$ has $\alpha_j \leq aP+(a+1)m$, so by (a) $\min\alpha \leq aP+(a+1)m$, and the same for β. If $\min\alpha+\min\beta > a(P+\Lambda)+2(a+1)m$, both minima exceed $a\Lambda + (a+1)m$. A D level with $f_j \leq a$ has $\alpha_j \leq a\Lambda + (a+1)m$ and an H level with $g_j \leq a$ has $\beta_j \leq a\Lambda + (a+1)m$, so every D level has $f_j \geq a+1$ and every H level $g_j \geq a+1$, contradicting (c). □

Theorem 2.2 [new; Theorem] (*the alternating layout*). Take the rotational tournament $j \to j+i$ $(1 \leq i \leq k)$ on $\mathbb{Z}_{2k+1}$ and the typing $\tau_j = H$ for j even, $\tau_j = D$ for j odd $(0 \leq j \leq 2k$; the levels $2k$ and 0 are both of type $H)$. Then $f_j = \lceil k/2 \rceil$ for $0 \leq j \leq k$, and for $j = k+t$ $(1 \leq t \leq k)$

$$f_j = \begin{cases} k/2 \text{ (t even, H)} & k/2+1 \text{ (t odd, D)} & k \text{ even}, \\ (k-1)/2 \text{ (t odd, H)} & (k+1)/2 \text{ (t even, D)} & k \text{ odd}; \end{cases}$$

and $g_j = f_{2k-j}$. Hence (for odd k using $(*)$)

$$k \text{ even}: \quad \min\alpha = \min\beta = \tfrac{k}{2}(\Lambda+m); \qquad k \text{ odd}: \quad \min\alpha = \tfrac{k-1}{2}P + \tfrac{k+1}{2}m, \quad \min\beta = \tfrac{k-1}{2}\Lambda + \tfrac{k+1}{2}m,$$

For k even both minima attain the bounds of Theorem 2.1; for k odd $\min\alpha$ and $\min\alpha + \min\beta$ do.

Proof. For $j \leq k$ the targets $j+1, \ldots, j+k$ need no reduction mod $2k+1$, and exactly $\lceil k/2 \rceil$ of them have the parity opposite to j. For $j = k+t$ the targets are $j+1, \ldots, 2k$ ($k-t$ of them, $\lceil (k-t)/2 \rceil$ of the other parity) and $0, \ldots, t-1$, of which $\lfloor t/2 \rfloor$ (if j is even) or $\lceil t/2 \rceil$ (if j is odd) have the other parity; the four cases give the values stated. The reflection $j \mapsto 2k-j$ preserves types and reverses every transition, so $g_j = f_{2k-j}$. For k even the minimum a of f is attained at D levels (odd $j \leq k$), so $\min\alpha = a\Lambda + am$; the minimum of g is attained at H levels, where friendly incoming sets are lines: $\min\beta = a\Lambda + am$. For k odd, $f_j = a$ only at H levels and every D level has $f_j = a+1$, whose $\alpha_j = (a+1)\Lambda + am \geq aP + (a+1)m$ by $(*)$; so $\min\alpha = aP + (a+1)m$; $g_j = a$ only at H levels, where $\beta_j = a\Lambda + (a+1)m$. □

So, under $(*)$, with friendly point sets of size P, friendly line sets of size Λ and non-friendly sets of size m, the alternating layout maximizes $K_{\mathcal{A}} + K_{\mathcal{B}}$ (and for k even each of them) over all regular tournaments and typings. Its slot counts are

$$K_{\mathcal{B}}(q,k) = K_{\mathcal{A}}(q,k) = \tfrac{k}{2}(\Lambda+m) \text{ (k even)}, \qquad K_{\mathcal{B}} = \tfrac{k-1}{2}P + \tfrac{k+1}{2}m, \quad K_{\mathcal{A}} = \tfrac{k-1}{2}\Lambda + \tfrac{k+1}{2}m \text{ (k odd)}.$$

For $k = 1$ this is m on both sides, `D7_HEXAGON.md` Corollary 9.1.

The layouts of the builds [Computer-checked against the builders' logs]. The free slots per level, in level order:

k	types	α_j	β_j
2	$HDHDH$ (alternating)	P+m, Λ+m, P+m, 2Λ, P+m	Λ+m, 2P, Λ+m, P+m, Λ+m
3	$HHDHHDD$	P+2m, P+2m, 2Λ+m, 2P+m, 2P+m, 2Λ+m, 2Λ+m	2Λ+m, 2Λ+m, 2P+m, Λ+2m, Λ+2m, 2P+m, 2P+m
4	$HHDHDHDHD$ (alternating, rotated)	2P+2m, 2P+2m, 2Λ+2m, 2P+2m, 2Λ+2m, 2P+2m, 3Λ+m, 2P+2m, 3Λ+m	2Λ+2m, 2Λ+2m, 3P+m, 2Λ+2m, 3P+m, 2Λ+2m, 2P+2m, 2Λ+2m, 2P+2m

The $k = 3$ layout of the builds is not the alternating one but has the same minima. With the values of each build these vectors equal, entry by entry, the vectors printed by `typed_build.py` (`d7thm/cells.out`).

Computer-checked. `d7thm/slots.py`: for $k \leq 8$ every typing of the rotational tournament (the optimum is the one of Theorem 2.2, numerically for $q = 11$ with $m = 768$, $q = 13$ with $m = 792$, $q = 16, 101, 1009$); every labelled regular tournament on 5 and 7 levels (24 and 2,640) with every typing; SAT (CaDiCaL) on $9, 11, 13$ levels finds no regular tournament and typing with $f_j, g_j \geq a$ everywhere, $f_j \geq a+1$ on D levels and $g_j \geq a+1$ on H levels (one combined instance; (b) and (c) themselves are proved above). Independently squad card c6: all regular tournaments on $3, 5, 7$ levels, 200,000 random ones on 9 and 11 levels, annealing, and the closed forms of Theorem 2.2 for $k \leq 40$. Card c6 also found that without $(*)$ the value of $\min\alpha$ for odd k changes (the D levels become the minimum); this happens only for k far above q (for instance $k \geq 27$ at $q = 11$), outside the range of the bound.

3. The bound

Theorem 3.1 [new; Theorem]. Let q be a prime power, q odd or $q \geq 4$, let $1 \leq k \leq q-1$, and let $K_{\mathcal{B}}(q,k)$, $K_{\mathcal{A}}(q,k)$ be the slot counts of Theorem 2.2 with $P = P(q)$, $\Lambda = \Lambda(q)$ and $m = q^2+1$. If an $\mathrm{HG}(K_1, K_2; q-k)$ exists with $K_1 \leq K_{\mathcal{B}}$ and $K_2 \leq K_{\mathcal{A}}$, then

$$n_b(q+1+k,\, 7) \;\geq\; 2(2k+1)N(q) + K_1 + K_2.$$

Proof. Take $2k+1$ copies of the incidence graph of $H(q)$, the alternating layout of Theorem 2.2, the identity on friendly transitions with freed sets $R_H = \mathcal{P}_S$ (source H) and R_L (source D) of Corollary 1.8, and on non-friendly transitions the local duality σ ($H \to H$) or σ^{-1} ($D \to D$) of Proposition 1.9 with the q^2+1 freed elements. Every friendly transition is repaired by Theorem B, every other one by Theorem C; $(*)$ holds since $k \leq q-1$ (§2), so Theorem 2.2 gives the slot counts. Attach K_1 hubs of colour $\mathcal{B}$ and K_2 of colour $\mathcal{A}$ to free slots in any way (one slot per level each; unused slots stay free) and use the given graph as hub graph. Theorem 0.1 applies. □

The same holds with m replaced by the size of any non-friendly set satisfying Theorem C, provided $(*)$ holds; and for any generalized hexagon of order (q,q) with a spread, an exact whisker set with the property of Theorem 1.7, and a partial ovoid of a size m satisfying $(*)$ (for instance $m \leq q^2+1$), by the same proof.

Corollary 3.2 [new; Theorem when the hub graph exists]. With full slots ($K_1 = K_{\mathcal{B}}$, $K_2 = K_{\mathcal{A}}$) Theorem 3.1 gives $n_b(q+1+k,7) \geq F_k(q)$, where

$$F_k(q) = 2(2k+1)(q+1)(q^4+q^2+1) + \begin{cases} k\,(q^4+2q^2+q+1), & k \text{ even}, \\ \frac{k-1}{2}\,(2q^4+q^3+q^2+2q+1) + (k+1)(q^2+1), & k \text{ odd}. \end{cases}$$

In particular

$$F_2(q) = 10N(q)+2q^4+4q^2+2q+2, \quad F_3(q) = 14N(q)+2q^4+q^3+5q^2+2q+5, \quad F_4(q) = 18N(q)+4q^4+8q^2+4q+4.$$

The gain over the level-graph value $2(2k+1)N(q)$ (the 2013 table value for the pairs of §0) is $kq^4 + O(q^2)$ for k even and $(k-1)q^4 + \frac{k-1}{2}q^3 + O(q^2)$ for k odd; the earlier general statement, `D7_HEXAGON.md` Corollary 9.3 ($k=2$), gave $2(q+2)(q^2+1) \approx 2q^3$, and none was stated there for $k \geq 3$ except for $q = 3^h$ (Corollary 9.2). With a larger non-friendly set m the formula changes by $k(m-q^2-1)$ for k even and $(k+1)(m-q^2-1)$ for k odd.

Computer-checked (`d7thm/cells.py`). The closed forms agree with the slot formulas for $q \leq 37$, $k \leq 6$.

4. Hub graphs

The bound needs $\mathrm{HG}(K_{\mathcal{B}}, K_{\mathcal{A}}; q-k)$ with $K \approx \lfloor k/2 \rfloor q^4$ hubs per side and degree $q-k$. The Moore-type bound for same-colour distance 6 is about $(q-k)^6$, so there is room, but the explicit hub graph of `D7_HEXAGON.md` (Lemma 7.1, a plane with pendant trees) has only about $\frac{4}{27}(q-k)^4$ vertices per colour. Write $K_0(s) = (s+1)(s^4+s^2+1)$ for the number of points of a hexagon of order (s,s).

Lemma 4.1 (*blown-up hexagon*) [standard; Theorem]. Duplicating elements of the split Cayley hexagon is Delorme's construction [De85a, §8(b)]; this is its uniform version, and the projection argument is standard. Let s be a prime power and $c \geq 1$. Replace every point and every line of $H(s)$ by between 1 and c copies, and join two copies when the originals are incident. The result is an $\mathrm{HG}(K_1, K_2; c(s+1))$ for any chosen $K_1, K_2 \in [K_0(s), cK_0(s)]$, and any two of its vertices of different colours are at distance at most 5.

Proof. A path between copies of $x \neq y$ projects to a walk from x to y of the same length, and a geodesic from x to y lifts to copies, so the distance is $d(x,y)$; two copies of one element are at distance 2. A copy of x has degree $\sum_{y \;\mathrm{I}\; x} t_y \leq c(s+1)$. □

Lemma 4.2 (*sums*) [new; Theorem]. Let $Y_1, \ldots, Y_r$ be bipartite graphs with colour classes (A_i, B_i), each with all same-colour distances at most 6, and for $i < r$ also all different-colour distances at most 5. Suppose $|A_j| \leq |B_i|$ and $|B_j| \leq |A_i|$ for $i < j$. For every $i < j$ add an injective map $A_j \to B_i$ and an injective map $B_j \to A_i$, as edges. The result is an $\mathrm{HG}(\sum |A_i|, \sum |B_i|;\ \max_i \Delta(Y_i) + r - 1)$.

Proof. The new edges join the colours A and B, so the graph is bipartite with classes $\bigcup A_i$, $\bigcup B_i$. A vertex of Y_i gains one edge to each Y_j, $j < i$, and at most one from each Y_j, $j > i$. Same-colour pairs inside one Y_i are at

distance at most 6 in Y_i. For $x \in A_j$ and $z \in A_i$ with $i < j$, x has a neighbour $b \in B_i$, and $d_{Y_i}(b, z) \le 5$ since $i < r$; so $d(x, z) \le 6$. The same for B. □ Lemma 4.2 is a compound in the sense of [GF85] (copies of graphs joined by matchings along a tournament); the new point is the order condition, with different-colour distance ≤ 5 in every part but the last, which makes every same-colour pair reach distance 6 through an earlier part.

Lemma 4.3 (`D7_HEXAGON.md`, Lemma 7.1, with the colours sized separately) [restatement; Theorem]. Let s be a prime power, $n = s^2 + s + 1$, $t = \delta - s - 1 \ge 0$, $u = \delta - 1$. Attach to the incidence graph of $\mathrm{PG}(2, s)$ c_P pendant vertices at the points and c_L at the lines (at most t per core vertex), and g_P, g_L pendant vertices below these: g_P below the c_P pendants at points, g_L below the c_L at lines (at most u per vertex). This is an $\mathrm{HG}(n + c_L + g_P,\ n + c_P + g_L;\ \delta)$. *Proof.* The distance argument of Lemma 7.1 does not use the numbers of pendant vertices. □ (Different-colour distances can be 7 here, so such a part can only be the last part Y_r in Lemma 4.2; squad card c5 confirmed this with a negative control.)

Theorem 4.4 [new; Theorem; (b) computer-assisted: the decompositions are found and verified by a program]. Let $k \ge 2$. An $\mathrm{HG}(K_{\mathcal{B}}(q,k), K_{\mathcal{A}}(q,k); q-k)$ (full slots of Theorem 2.2, $m = q^2 + 1$) exists

(a) for every prime power $q \ge 300k$;

(b) for $2 \le k \le 12$, for every prime power $q \ge q_k$, where

k	2	3	4	5	6	7	8	9	10	11	12
q_k	16	23	23	29	23	31	29	37	32	43	41

These are the smallest q from which our sums of Lemmas 4.1–4.3 reach the full slots: `d7thm/hubopt.py` and a second, independently written search (`research/referee/d7thm_checks/mysearch.py`, branch `claude/referee-d7-theorems-8og1h3`) agree at every (q, k) with $k \le 12$ and $q < q_k$, but neither is exhaustive over size assignments. Other hub graphs may do better. They are not monotone in k.

Proof. (a) Let $q \ge 300k$ (so $q \ge 600$). The slot counts satisfy $q^4 < \Lambda \le K_{\mathcal{A}} \le K_{\mathcal{B}}$, $K_{\mathcal{B}}/K_{\mathcal{A}} \le P/\Lambda \le 1 + 1/q$ and $K_{\mathcal{B}} \le kP \le k(q+1)^4$. Put $x = K_{\mathcal{A}}^{1/5} - 1$ ($> q^{4/5} - 1 > 160$). By Nagura's theorem [Na52] (for $y \ge 25$ there is a prime in $(y, \frac{6}{5}y)$) there is a prime s with $x/1.2 < s < x$. Then $K_0(s) < K_0(x) < (x+1)^5 = K_{\mathcal{A}}$, and, with $K_0(y) \ge y^5$, $x^5 \ge K_{\mathcal{A}}(1 - 5K_{\mathcal{A}}^{-1/5})$ (Bernoulli) and $K_{\mathcal{A}}^{-1/5} < q^{-4/5} \le 0.006$,

$$\frac{K_{\mathcal{B}}}{K_0(s)} < \frac{1.2^5 K_{\mathcal{B}}}{x^5} \le \frac{2.4884\,(1 + 1/q)}{1 - 0.03} < 3.$$

So Lemma 4.1 with $c = 3$ gives an $\mathrm{HG}(K_{\mathcal{B}}, K_{\mathcal{A}}; 3(s+1))$, and $3(s+1) < 3(x+1) = 3K_{\mathcal{A}}^{1/5} \le 3k^{1/5}(q+1)^{4/5} = 3(q+1)(k/(q+1))^{1/5} \le 3 \cdot 300^{-1/5}(q+1) < 0.959\,(q+1) \le q - k$, the last step because $0.041(q+1) \ge 12.3k \ge k+1$. (b) For $q \le 200$ the program `d7thm/hubopt.py` (an enumeration of multisets of parts with bounds) writes for each (q, k) a decomposition into parts of Lemmas 4.1–4.3, combined by Lemma 4.2, and verifies it (sizes add up, the order condition, the degrees, the pendant-tree parameters, which `hubopt.out` prints); each line can be checked by hand. Examples: $(q, k) = (16, 2)$, $K = 66{,}065$: $H(8) + H(7) + H(5)$ and two copies of $H(4)$ blown up to 2,730 and 2,372 (degrees $9+4$, $8+4$, $6+4$, $10+4 \le 14$; this is the certificate that `build_cert.py` builds and `cert_checks.out` checks); $(q, k) = (23, 3)$: $H(11) + H(9) + H(8)$, the last blown up to 49,506 and 37,867. Below the full slots the same search gives the hub graphs of §5.2, for instance $(13, 4)$: $H(7) + H(7)$, two hexagons of order 7 joined by a perfect matching from the points of each to the lines of the other (39,216 per colour, degree $8 + 1 = 9$). For $200 < q \le 2 \cdot 10^5$ a single blow-up with $c \le 2$ suffices (Lemma 4.1 alone; `d7thm/hubcap.py` finds s and c for every such q); for $q > 2 \cdot 10^5 \ge 300k$ use (a). □

Computer-checked. `d7thm/build_cert.py` builds the hub graph of a certificate (hexagons from `research/d10D7/hexgen2.c`, random injections) and `hgcheck edges` checks it: $(q, k) = (16, 2)$, $\mathrm{HG}(66065, 66065; 14)$ (the one behind $(19, 7)$), $(17, 2)$, $\mathrm{HG}(84117, 84117; 15)$, and $(23, 3)$, $\mathrm{HG}(293092, 281453; 20)$: no same-colour pair at distance more than 6, degrees as claimed (`d7thm/cert_checks.out`).

Computer-checked hub graphs below q_k (`d7thm/hgcheck.c`, own code: bit-parallel breadth-first search from every hub; `d7thm/hgcheck_lab.out`, `d7thm/hgrun.out`). The lab's stored hub graphs, read from the solution files, re-checked standalone: $\mathrm{HG}(15541, 15541; 9)$ [(14,7)], $\mathrm{HG}(29535, 29535; 11)$ [(16,7)], $\mathrm{HG}(32356, 30327; 10)$

and HG(32356, 30158; 10) [(17,7)], HG(25308, 25308; 9) [(18,7), PSL(2,37)], HG(66065, 66065; 14) [(19,7)], HG(70163, 66322; 13) [(20,7)], and the Haar graph over PSL(2, 29) of job "conics", HG(12180, 12180; 8) [(15,7), `conics/d15D7/hub_psl29.txt`]: all VALID. The two cluster-assisted ones [(15,7) 16,152, (18,7) 34,440] are not hub graphs in this sense (their same-colour pairs are joined through the levels; in the hub graph alone 452,256 and 654,360 ordered same-colour pairs are at distance more than 6, the second number as in `conics2/README.md`); for them (HG) is covered by the all-pairs checks of the whole graphs. New random graphs (`hgcheck rand`, up to four seeds, aimed 2-switch repair when $K \le 60{,}000$) at the slot counts of Theorem 2.2 with $m = q^2 + 1$, for the (q, k) with $11 \le q < q_k$, $k \le 5$ (ordered same-colour pairs at distance more than 6 are "far pairs"; Haar graphs of $\mathbb{Z}_K$ were also tried and never covered):

(q, k)	$\delta = q - k$	$K_{\mathcal{B}}$	$K_{\mathcal{A}}$	$M(\delta)/K_{\mathcal{B}}$	result
$(11, 2)$	9	14,895	14,895	20.1	found (rand seed 2)
$(13, 2)$	11	28,913	28,913	38.4	found (rand seed 1)
$(11, 3)$	8	16,228	15,017	8.5	not found (last graph tried: 602,324 far pairs)
$(13, 3)$	10	31,112	29,083	19.2	not found (last graph tried: 44 far pairs)
$(16, 3)$	13	70,163	66,322	46.4	found (rand seed 1)
$(17, 3)$	14	89,032	84,407	58.7	found (rand seed 1)
$(19, 3)$	16	137,924	131,425	88.5	found (rand seed 1)
$(11, 4)$	7	29,790	29,790	1.9	not found (last graph tried: 270,820,464 far pairs)
$(13, 4)$	9	57,826	57,826	5.2	not found (last graph tried: 37,527,036 far pairs)
$(16, 4)$	12	132,130	132,130	14.7	not found (last graph tried: 13,824 far pairs)
$(17, 4)$	13	168,234	168,234	19.4	not found (last graph tried: 212 far pairs)
$(19, 4)$	15	262,126	262,126	30.9	found (rand seed 1)
$(11, 5)$	6	32,334	29,912	0.6	not found (last graph tried: 1,190,378,570 far pairs)
$(13, 5)$	8	62,054	57,996	2.2	not found (last graph tried: 1,152,049,654 far pairs)

(q,k)	$\delta = q-k$	$K_{\mathcal{B}}$	$K_{\mathcal{A}}$	$M(\delta)/K_{\mathcal{B}}$	result
$(16,5)$	11	140,069	132,387	7.9	not found (last graph tried: 43,492,232 far pairs)
$(17,5)$	12	177,774	168,524	11.0	not found (last graph tried: 5,026,754 far pairs)
$(19,5)$	14	275,486	262,488	19.0	not found (last graph tried: 11,012 far pairs)

We did not run $k = 5$ with $q \geq 23$ (more than 300,000 hubs per colour) or $k = 6$.

Heuristic. Let $M(\delta) = 1 + \delta(\delta-1) + \delta(\delta-1)^3 + \delta(\delta-1)^5$, the most vertices of one colour within distance 6 of a vertex in a graph of maximum degree δ. In the table above a random hub graph worked exactly when $M(\delta)/K_{\mathcal{B}} \geq 20$; at 19.0–19.4 ($(19,5)$, $(13,3)$, $(17,4)$) tens to thousands of far pairs were left, and below that random graphs are far off. Where they fail the lab used Haar graphs over $\mathbb{Z}_{673} \rtimes \mathbb{Z}_{24}$ and $\mathrm{PSL}(2,41)$ and paths through the levels, which reach fewer hubs than the slots (16,152 of 16,309 at $(15,7)$, 34,440 of 59,070 at $(18,7)$).

5. The cells

5.1 The fifteen built graphs

`d7thm/cells.py` recomputes, for every graph built in the jobs "conics" and "conics-2", the slot vectors of §2 from q, the type string, $|\mathcal{P}_S| = P(q)$, $|R_L|$ (q^4+q without S_0, q^4+q^2+q with it) and m, and the order $2(2k+1)N(q) + K_{\mathcal{A}} + K_{\mathcal{B}}$ from the hub counts. **Computer-checked**: all fifteen agree with the builders (`cells.out`): twelve of the fifteen builds have a make log on branch `claude/conics-2-diameter-7-ic0oat`, and all twelve slot vectors equal the printed ones (this includes the $(15,7)$ build with 16,152 hubs per side, whose `make_meta.log` is not compared by `cells.py`). The three builds of job "conics" (orders 1,802,400, 2,504,544, 4,081,072 for $(14,7)$, $(15,7)$, $(16,7)$) have no make log and are compared by order only; they agree. Every order equals the n of its solution file. The current graphs:

cell	q, types	m	hubs per side	hub graph	n	all-pairs check	the bound follows from
(14,7)	11, $HDHDH$	768	15,541 + 15,541	random	1,802,642	passed	Thm 3.1 + computer (set, hub graph)
(15,7)	11, $HHDHHDD$	768	16,152 + 16,152	Haar, clusters	2,512,488	passed	all-pairs check
(16,7)	13, $HDHDH$	792	29,535 + 29,535	random	4,081,410	passed	Thm 3.1 + computer (set, hub graph)
(17,7)	13, $HHDHHDD$	792	32,356 + 30,327	random	5,693,959	passed	Thm 3.1 + computer (set, hub graph)
(18,7)	13, $HHDHDHDHD$	792	34,440 + 34,440	Haar, clusters	7,309,092	passed	all-pairs check
(19,7)	16, $HDHDH$	257	66,065 + 66,065	random	11,316,940	9/16 ranges	**Theorem** ($= F_2(16)$)
(20,7)	16, $HHDHHDD$	257	70,163 + 66,322	random	15,795,219	9/24 ranges	Thm 3.1 + computer (hub graph) ($= F_3(16)$)

All use $\Lambda = q^4 + q^2 + q$ (with S_0). Hub graphs: "random" means random bipartite of degree $q - k$ (checked standalone, §4); "Haar, clusters" means a Haar graph over $\mathbb{Z}_{673} \rtimes \mathbb{Z}_{24}$ resp. $\mathrm{PSL}(2, 41)$ whose same-colour pairs are partly joined through the levels (`conics2/README.md` §5). The all-pairs states are those of `conics2/README.md` at 11:44 UTC.

At $(19, 7)$ and $(20, 7)$ the built graphs use exactly the sets of Theorem 3.1 ($m = q^2+1$, the elliptic section), so their orders are $F_2(16)$ and $F_3(16)$. For $(19, 7)$ every ingredient is proved: the sets (Corollary 1.8, Proposition 1.9), the repair of every transition (Theorems B and C), the slot counts (Theorem 2.2) and an $\mathrm{HG}(66065, 66065; 14)$ (Theorem 4.4: $H(8) + H(7) + H(5)$ and two copies of $H(4)$ blown up to 2,730 and 2,372 vertices per colour, joined by Lemma 4.2; this is the certificate that `d7thm/cert_checks.out` checks). So

$$n_b(19, 7) \;\geq\; 11{,}316{,}940$$

holds by proof, independently of the all-pairs check of the built graph (which uses a random hub graph and was not complete when job "conics-2" ended). For $(20, 7)$ the hub graph $\mathrm{HG}(70163, 66322; 13)$ is not covered by Theorem 4.4; it is computer-checked (the builder's graph, re-checked standalone by `hgcheck`, and a second random graph, seed 1, `hgrun.out`); with the proved equal-sided hub graph of 66,234 hubs per colour the bound is 15,791,202, as in §5.2. This is a statement about the theorems, not a record claim: the lab records values after the coordinator's own check.

A larger (18,7) graph [new; Theorem; all ingredients explicit; built and partly computer-checked]. At $(q, k) = (13, 4)$ the cluster-assisted Haar hub graph of the current build has 34,440 hubs per colour, while Lemma 4.2 gives $\mathrm{HG}(39216, 39216; 9)$ ($H(7) + H(7)$, §4). With the build's sets (slots 59,070 per colour) or those of Theorem 3.1 (slots 57,826, from Corollary 1.8, Proposition 1.9 and Theorem 2.2; no computer search is needed for this bound), Theorem 3.1 gives

$$n_b(18, 7) \;\geq\; 18 \cdot 402{,}234 + 2 \cdot 39{,}216 \;=\; 7{,}318{,}644,$$

9,552 more than the value 7,309,092 of branch `claude/conics-2-diameter-7-ic0oat` (not on master). We built it with `conics2/typed_build.py` (same hexagon file, types $HHDHDHDHD$, the 792-point mirrored set, S_0 included; the hub graph from `d7thm/build_cert.py`): the builder re-checks every hypothesis (segment and dual segment sets, the local duality at all 792 points, same-colour hub distances at most 6, degrees 15 to 18) and writes n = 7,318,644, 65,612,918 edges, edge-list sha256 `c80d9c0e…ed1a` (full hash in `d7thm/d18D7/README.md`); the solution file `research/frontier/sols/d18D7_hex13_typedwhisk_h7h7_n7318644.sol` rebuilds it byte for byte (`typed_build.py build`). **Computer-checked:** (i) its level part (all edges between level vertices, 64,554,086 of them) is identical to that of the lab's 7,309,092 graph, rebuilt from its solution file (sha256 `314d5db9…`, as recorded); (ii) `apsp_fast` (`conics/apsp_fast.c`) from all 78,432 hubs: bipartite, degrees 15 to 18, every hub within distance 7 of every vertex (`RESULT VALID d=18 D=7 n=7318644` for the sources 7,240,212 to 7,318,643); (iii) `apsp_fast` from three ranges of level vertices (sources 0–262,143 in level 0, 804,468–1,066,611 in level 1, 6,978,068–7,240,211 in level 8; 786,432 sources, 10.9% of the level vertices): all `RESULT VALID`, eccentricity at most 7. Pairs of level vertices are within distance 7 inside two levels by Theorems B and C, whose hypotheses the builder checked; a full all-pairs check (about 20 hours on 4 cores here) was not run and is left to the coordinator (`d7thm/d18D7/`).

5.2 The cells $\Delta = 14, \ldots, 26$

For each Δ, the best (q, k) with $q + 1 + k = \Delta$; "level graphs" is $\max 2(2k + 1)N(q)$, the 2013 table value for $\Delta \leq 16$ and the lab's beyond-table value for $\Delta = 17, \ldots, 20$ (`research/frontier/beyond/BEYOND_TABLE.md`, branch `claude/new-cells-beyond-etos3t`). $F_k(q)$ is Corollary 3.2 with $m = q^2 + 1$; its hub graph is "proof" (Theorem 4.4), "computer" (`hgcheck`) or "open". "Proved" is the value of Theorem 3.1 with the hub graph that `d7thm/hubopt.py` finds below the slots (a sum by Lemma 4.2, shown in the last column; full slots where Theorem 4.4 applies). The search is not proved optimal, so these are values found, not the best the lemmas allow.

cell	level graphs	(q,k)	$F_k(q)$	hub graph at full slots	proved	at (q,k): hubs	hub graph of the proved value
(14,7)	1,771,560	$(11,2)$	1,801,350	computer	1,797,726	$(11,2)$: 13,083 + 13,083	H(5) + H(5) + H(5) + H(4)
(15,7)	2,480,184	$(11,3)$	2,511,429	open	2,503,620	$(11,3)$: 11,718 + 11,718	H(5) + H(5) + H(5)
(16,7)	4,022,340	$(13,2)$	4,080,166	computer	4,079,910	$(13,2)$: 28,785 + 28,785	H(7) + H(5) + H(5) + H(4)
(17,7)	5,631,276	$(13,3)$	5,691,471	open	5,686,116	$(13,3)$: 27,420 + 27,420	H(7) + H(5) + H(5)
(18,7)	7,240,212	$(13,4)$	7,355,864	open	7,318,644	$(13,4)$: 39,216 + 39,216	H(7) + H(7)
(19,7)	11,184,810	$(16,2)$	11,316,940	proof	11,316,940	$(16,2)$: 66,065 + 66,065	H(8) + H(7) + H(5) + H(4)x2 + H(4)x2
(20,7)	15,658,734	$(16,3)$	15,795,219	computer	15,791,202	$(16,3)$: 66,234 + 66,234	H(8) + H(7) + H(5) + H(5) + H(4)
(21,7)	21,120,372	$(17,3)$	21,293,811	computer	21,290,219	$(17,3)$: 85,440 + 84,407	H(8) + H(8) + H(5) + H(5) + H(4)x2
(22,7)	27,154,764	$(17,4)$	27,491,232	open	27,476,636	$(17,4)$: 160,936 + 160,936	H(9) + H(8) + H(8) + H(7)
(23,7)	36,591,240	$(19,3)$	36,860,589	computer	36,857,870	$(19,3)$: 135,205 + 131,425	H(9) + H(8) + H(7) + H(5)x2 + H(5)
(24,7)	47,045,880	$(19,4)$	47,570,132	computer	47,566,386	$(19,4)$: 260,253 + 260,253	H(9) + H(9) + H(9) + H(8) + H(7) + H(5)
(25,7)	57,500,520	$(19,5)$	58,038,494	open	58,013,214	$(19,5)$: 256,347 + 256,347	H(9) + H(9) + H(9) + H(8) + H(7)
(26,7)	67,955,160	$(19,6)$	68,741,538	open	68,663,784	$(19,6)$: 354,312 + 354,312	H(11) + H(11)

At $(15,7)$ the lab's Haar graph over $\mathrm{PSL}(2,29)$ (`CONICS.md` §3.4; re-checked above: same-colour distances at most 6 in the hub graph alone) fits the slots of Theorem 2.2 with $m = q^2+1$ as well, so Theorem 3.1 with that computer-checked hub graph gives $14N(11) + 2 \cdot 12{,}180 = 2{,}504{,}544$.

At $(18,7)$ the proved value, 7,318,644, exceeds the lab's built 7,309,092 (§5.1). The proved values exceed the level-graph values at every cell $\Delta = 14, \ldots, 26$. For $k \geq 3$ the lab's notes had no general statement except for $q = 3^h$ (`D7_HEXAGON.md`, Corollary 9.2; §9 there says none covers $(15,7)$). The values with computer-found ingredients (the lab's sets $m = 768, 792$ and hub graphs) are the built values of §5.1 and are larger at $(14,7)$–$(17,7)$; at $(18,7)$ the explicit hub graph $H(7) + H(7)$ beats the computer-found one.

6. Summary: what is new

result	label	status
Theorem 1.4: lines meeting v, whiskers and the spread lines missed by them form a dual segment set, for any hexagon, spread and whisker set; maximality criterion	new in this form (special cases: `CONICS.md` Lemma 3.1, `conics2` Prop. 1(c),(d))	Theorem, coordinate-free

result	label	status
Proposition 1.5: in a dual segment set whose lines meet only in spread points, at most q^4+q lines are off the spread; $\mathcal{L}_v \cup W$ attains it	new	Theorem
Lemma 1.6: an exact whisker set misses at most q^3-q^2 spread lines, exactly q^2 under the "all or nothing" property	new	Theorem
Theorem 1.7, Corollary 1.8: $\lvert R_H\rvert=(q+1)(q^3+1)$, $\lvert R_L\rvert=q^4+q^2+q$, R_L maximal, every odd q and even $q\geq 4$	restatement of `conics2` Prop. 1 with proof	Theorem; Computer-checked $q=3,\dots,16$ (squad c1–c3, builder; $q=2$ checked to have no zero-free G)
Theorem 2.1: for every regular tournament and typing, the hub counts per side are at most $\lfloor k/2\rfloor$ friendly sets, with the exact second-order term	new	Theorem; Computer-checked (all tournaments on ≤ 7 levels, SAT to 13 levels, squad c6)
Theorem 2.2: the alternating layout attains Theorem 2.1 for every k; closed forms	new	Theorem; Computer-checked $k\leq 40$
Theorem 3.1, Corollary 3.2: $n_b(q+1+k,7)\geq F_k(q)$ given the hub graph	new (for $k=2$ it improves `D7_HEXAGON.md` Cor. 9.3 from $\approx 2q^3$ to $\approx 2q^4$ above the level graph; nothing was stated for $k\geq 3$, $q\neq 3^h$)	Theorem
Lemma 4.2: sums of hub graphs; Lemma 4.1: blow-ups; Lemma 4.3: `D7_HEXAGON.md` Lemma 7.1 with the colours sized separately	4.2 new; 4.1 elementary; 4.3 restatement	Theorem; Computer-checked (squad c5)
Theorem 4.4: the hub graphs at full slots exist for $q\geq q_k$ ($q_2=16$, $q_3=q_4=23$, …, $q_k\leq 300k$)	new	Theorem; (b) computer-assisted (certificates by program for $q\leq 200$)
$n_b(19,7)\geq 11{,}316{,}940$ and $n_b(18,7)\geq 7{,}318{,}644$ (above the lab's 7,309,092) by proof; proved values above the level graphs at $(14,7)$–$(26,7)$ (§5.2)	new consequence	Theorem; the $(18,7)$ graph built, checked from every hub, full all-pairs check open
The formulas reproduce all fifteen built graphs; the lab's hub graphs re-checked standalone; new random hub graphs for $q<q_k$	—	Computer-checked
Random hub graphs fail at $(q,k)=(11,3),(13,4)$; larger dual segment sets than q^4+q^2+q for $q\geq 5$; a coordinate-free description of the q^2 lines with $\nu=0$	—	open; Heuristic remarks in §1.3 and §4

The theorems explain the values built by the two jobs (at $q=16$ exactly) and give the same kind of bound for every q and every k, with an explicit hub graph wherever $q\geq q_k$. One value is raised: at $(18,7)$ the explicit hub graph beats the computer-found one, giving 7,318,644 (a candidate until the coordinator's all-pairs check).

Review. A separate referee session re-derived the numerical steps (the Nagura chain, the closed forms, every table

entry, all certificates of `hubopt.out` with its own checker, and Theorem 2.1 on all regular tournaments with at most 7 levels) and found no error in the proofs; it found that the first certificate search was far from optimal (now `hubopt.py`, which produced the $(18, 7)$ hub graph) and several wording problems, all corrected here.

Appendix: files

In `research/synthetic/d7thm/` (Python 3; `pip install python-sat` for `slots.py`; C with OpenMP for `hgcheck.c`; commands in `d7thm/README.md`):

- `slots.py` → `slots.out`: §2 (all typings of the rotational tournament for $k \leq 8$, all regular tournaments on 5 and 7 levels, SAT for 9–13 levels).
- `hubcap_lib.py`, `hubcap.py` → `hubcap.out`: Lemma 4.1 coverage for $q \leq 2 \cdot 10^5$ and the inequalities of Theorem 4.4(a); `hubopt.py` → `hubopt.out`: the certificates of Theorem 4.4(b) and of §5.2 for $q \leq 200$ (one line per (q, k), pendant-tree parameters included); `build_cert.py`: the hub graph of a certificate as an edge list.
- `d18D7/`: the $(18, 7)$ graph of §5.1 (builder log, level-part comparison, the check from every hub); its solution file is `research/frontier/sols/d18D7_hex13_typedwhisk_h7h7_n7318644.sol`.
- `hgcheck.c`: standalone check of a hub graph (same-colour distances by bit-parallel BFS from every vertex); modes `sol` (read a solution file), `rand` (random bipartite graph, optional aimed 2-switch repair), `haar` (Haar graphs of $\mathbb{Z}_K$). `hgcheck_lab.out`: the lab's hub graphs; `hgrun.py` → `hgrun.out`: random hub graphs at the slot counts of Theorem 2.2 for $q < q_k$.
- `cells.py` → `cells.out`: §5.1 and the closed forms of Corollary 3.2; `table.py` → `table.out`: §5.2.

Squad round `research/squads/d7thm-r1/`: `BRIEF.md`, cards `c1`–`c6` (code and logs), `scoreboard.jsonl`; logged in `research/squads/LOG.md`. The PDF `D7_THEOREMS.pdf` is made from this file by pandoc and tectonic (`d7thm/make_pdf.sh`).

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